

TrES-1b

R-0111

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_241111 · created 2026-07-16T04:11:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460625.6232 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.051
Transit depth (Rp/Rs)^2	2.7 +/- 1.68 [%]
Orbital Inclination (inc)	89.11 +/- 1.41
Airmass coefficient 1 (a1)	1.126 +/- 0.04
Airmass coefficient 2 (a2)	-0.045 +/- 0.016
Scatter in the residuals of the lightcurve fit is	3.53 %
Best Comparison Star	#3 - [101, 244]
Optimal Aperture	2.56
Optimal Annulus	8.1
Transit Duration (day)	0.0971 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.051 vs 0.1358 (deviation 0.55σ)
Duration fit vs published	0.0971 d vs 0.1042 d (deviation 1.33σ)
Mid-time O–C	-7.6 min
Depth SNR	3.2
Residual scatter	3.53 %

Light curve

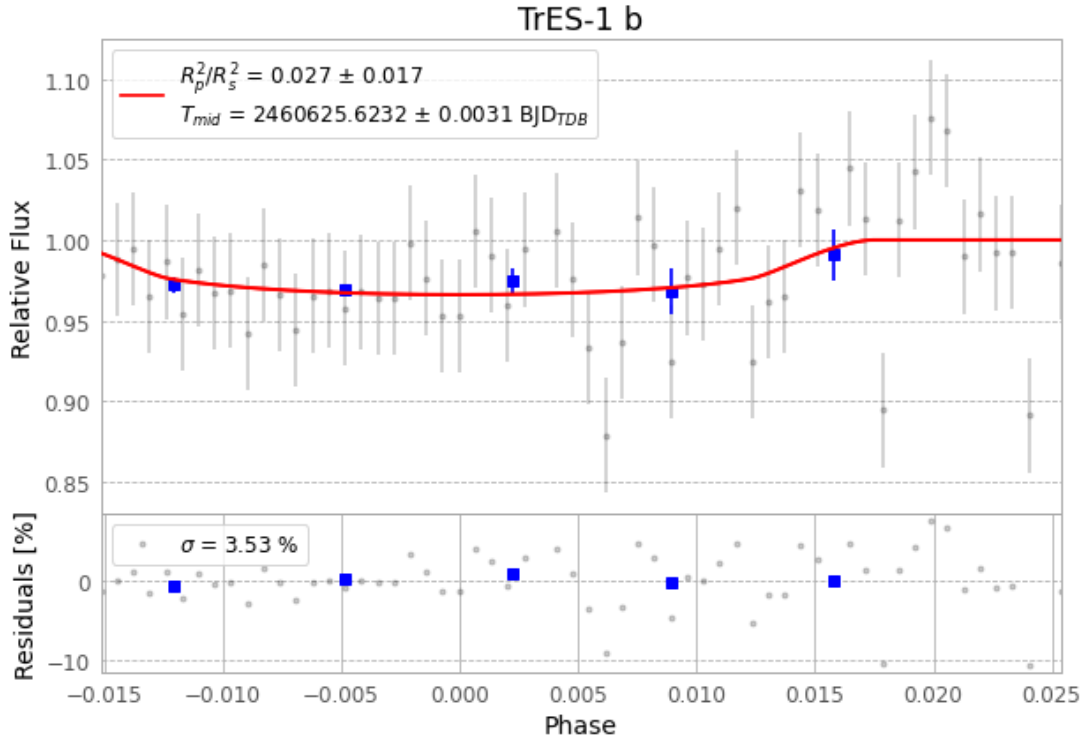


Figure 1 — FinalLightCurve_TrES-1 b_2024-11-10.png (SHA-256 ae6d0060177f1e75...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 285], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ae1f53fb2b4ac2a7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

71 FITS frames (47 MB), TRES-1241111015115.FITS → TRES-1241111052115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-241111.log (e2c01f90260a...), FinalLightCurve_TrES-1 b_2024-11-10.png (ae6d0060177f...), FinalLightCurve_TrES-1 b_2024-11-10.csv (5522a1f376f5...)

Compute resources

Wall time 225.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 04:11Z.